With this very simple example, you have performed a Bayesian analysis. Each new piece of information affected the original probability, and that is a Bayesian inference.

Bayesian analysis is an attempt to incorporate all available information into a process for making inferences, or decisions, about the underlying state of nature. Colleges and universities use Bayes's theorem to help their students study decision making. In the classroom, the Bayesian approach is more popularly called the decision tree theory; each branch of the tree represents new information that, in turn, changes the odds in making decisions. "At Harvard Business School," explains Charlie Munger, "the great quantitative thing that bonds the first-year class together is what they call decision tree theory. All they do is take high school algebra and apply it to real life problems. The students love it. They're amazed to find that high school algebra works in life."[11]

As Charlie points out, basic algebra is extremely useful in calculating probabilities. But to put probability theory to practical use in investing, we need to look a bit deeper at how the numbers are calculated. In particular, we need to pay attention to the notion of frequency.

What does it mean to say that the probability of guessing heads on a single coin toss is one in two? Or that the probability that an odd number will appear on a single throw of the dice is one in two? If a box is filled with 70 red marbles and 30 blue marbles, what does it mean that the probability is three in 10 that a blue marble will be picked? In all these examples, the probability of the event is what is referred to as a frequency interpretation, and it is based on the law of averages.

If an uncertain event is repeated countless times, the frequency of the event's occurrence is reflected in its probability. For example, if we toss a coin 100,000 times, the number of times that are expected to be heads is 50,000. Note I did not say it would be equal to 50,000. The law of large numbers says the relative frequency and the probability need be equal only for an infinite number of repetitions. Theoretically, we know that, in a fair coin toss, the chance of getting heads is one in two, but we will never be able to say the chance is equal until an infinite number of tosses have passed.